Integration & Applications

Basic antiderivatives (the power rule)

1 Find $\int x^4\,dx$.

2 Find $\int 3x^2\,dx$.

3 Find $\int \sqrt{x}\,dx$.

4 Find $\int \frac{1}{x^3}\,dx$.

Antiderivatives of polynomials

5 Find $\int (4x^3 - 6x + 5)\,dx$.

6 Find $\int (2x + 7)\,dx$.

7 Find $\int (x^2 - 4x + 3)\,dx$.

Antiderivatives of trigonometric and exponential functions

8 Find $\int \cos x\,dx$.

9 Find $\int \sin x\,dx$.

10 Find $\int e^x\,dx$.

Evaluating definite integrals

11 Evaluate $\int_0^2 3x^2\,dx$.

12 Evaluate $\int_1^3 2x\,dx$.

13 Evaluate $\int_0^4 x\,dx$.

14 Evaluate $\int_{-1}^2 3x^2\,dx$.

Definite integrals with polynomial integrands

15 Evaluate $\int_{0}^{2}(x^2+1)\,dx$.

16 Evaluate $\int_{1}^{3}(2x^2-3x+1)\,dx$.

17 Evaluate $\int_{0}^{1}(6x^2-4x+2)\,dx$.

18 Evaluate $\int_{-2}^{1}(x^2-x)\,dx$.

Definite integrals of trigonometric and exponential functions

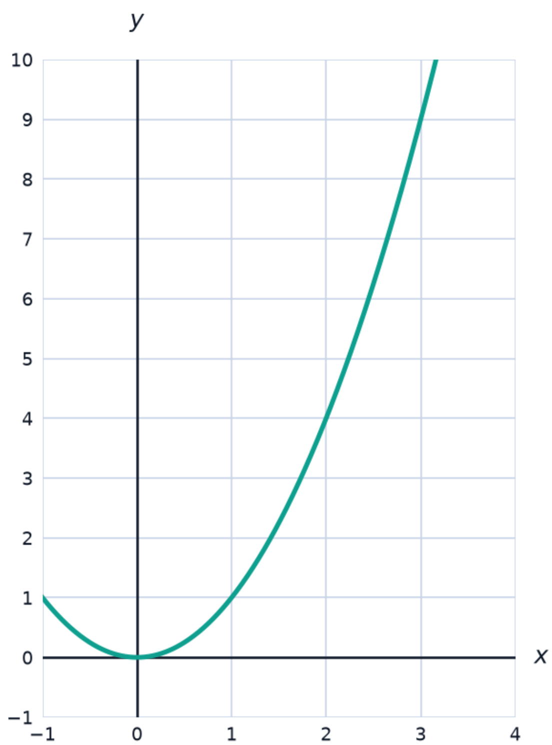
19 Evaluate $\int_{0}^{\pi/2}\cos x\,dx$.

20 Evaluate $\int_{0}^{\pi}\sin x\,dx$.

21 Evaluate $\int_{0}^{1}e^{x}\,dx$.

Area under a curve

22 Find the area under $y=x^2$ from $x=0$ to $x=3$, using a definite integral.



23 Find the area under $y=4-x^2$ from $x=-2$ to $x=2$.

24 Find the area under $y = 3\sqrt{x}$ from $x = 0$ to $x = 4$.

Signed area and regions below the axis

25 Evaluate $\int_0^2 (x-2)\,dx$ and explain what the sign of the result means.

26 Find the (positive) area between $y = x - 2$ and the x-axis, from $x = 0$ to $x = 2$.

Properties of definite integrals

27 Given $\int_1^4 f(x)\,dx = 10$, find $\int_1^4 3f(x)\,dx$.

28 Given $\int_0^2 f(x)\,dx = 5$ and $\int_2^5 f(x)\,dx = 7$, find $\int_0^5 f(x)\,dx$.

29 Given $\int_0^3 f(x)\,dx = 8$, find $\int_3^0 f(x)\,dx$.

Finding a function from its derivative

30 A function satisfies $f'(x) = 6x^2 - 4$ and $f(1) = 5$. Find $f(x)$.

31 A function satisfies $f'(x) = 3x^2 + 2x$ and $f(0) = 1$. Find $f(x)$.

32 A function satisfies $f'(x) = 4x - 1$ and $f(2) = 10$. Find $f(x)$.

Average value of a function

33 Find the average value of $f(x) = x^2$ on $[0, 3]$, using $\text{avg} = \frac{1}{b-a}\int_a^b f(x)\,dx$.

34 Find the average value of $f(x) = 2x + 1$ on $[1, 5]$.

Word problems: integration in context

35 This question has two parts. A particle moves along a line with velocity $v(t) = 6t - 3t^2$ (m/s) for $0 \le t \le 4$.

a Find the particle's displacement from $t = 0$ to $t = 2$, using $\int_0^2 v(t)\,dt$.

b Find the particle's displacement from $t = 0$ to $t = 4$, and explain why it is not simply twice the part (a) answer.

36 This question has two parts. Water flows into a tank at a rate of $r(t) = 20 - 2t$ liters per minute, for $0 \le t \le 10$.

a Write a definite integral for the total volume of water that flows in during the first 5 minutes.

b Evaluate the integral to find the volume.